

collaboration, be it for education, healthcare, or just office collaboration. Then there are use cases like ultra-rich multimedia content where we can use AR/VR for entertainment and gaming.”

Rajan S Mathews, director general, COAI, says, “With 5G, technologies such as autonomous driverless cars, smart transportation and logistics, smart farming, drone patrolling for road safety, holograms, and remote robotic healthcare are expected to enter our world and become more and more pervasive by the day. The technology is poised to open up a plethora of possibilities in terms of business models, better education, healthcare, smart cities, smart manufacturing, intelligent logistics, and overall enhanced lifestyles. The potential can be imagined by the fact that by 2026 there will be an estimated 619 billion dollars revenue potential for telecom operators.”

Yash Jethani, research manager, Regional Telecommunications Team, IDC Asia/Pacific, says, “For operators, new areas are clearly in education, healthcare, and government (public sector as IDC defines it) in the Covid environment, but in the long term, driverless connected cars, 4K/8K streaming, smart cities, and mixed reality/immersive videos are some interesting use cases for Asia/Pacific operators. India’s aggregate telecom spends on fixed data and fixed voice is thirty per cent and forty per cent, respectively, from B2C. B2C spend is even more predominant in the mobile data and mobile voice segments. Hence, for any future uptake in the market, the B2B focus also gives operators newer avenues to explore. With newer avenues, there will be numerous opportunities to capture under the government’s push with Digital India.”

### When can we expect 5G in India

As per the Organisation for Economic Cooperation and Development (OECD) committee on digital economic policy, 5G will not only amplify the country’s GDP but will also open up new

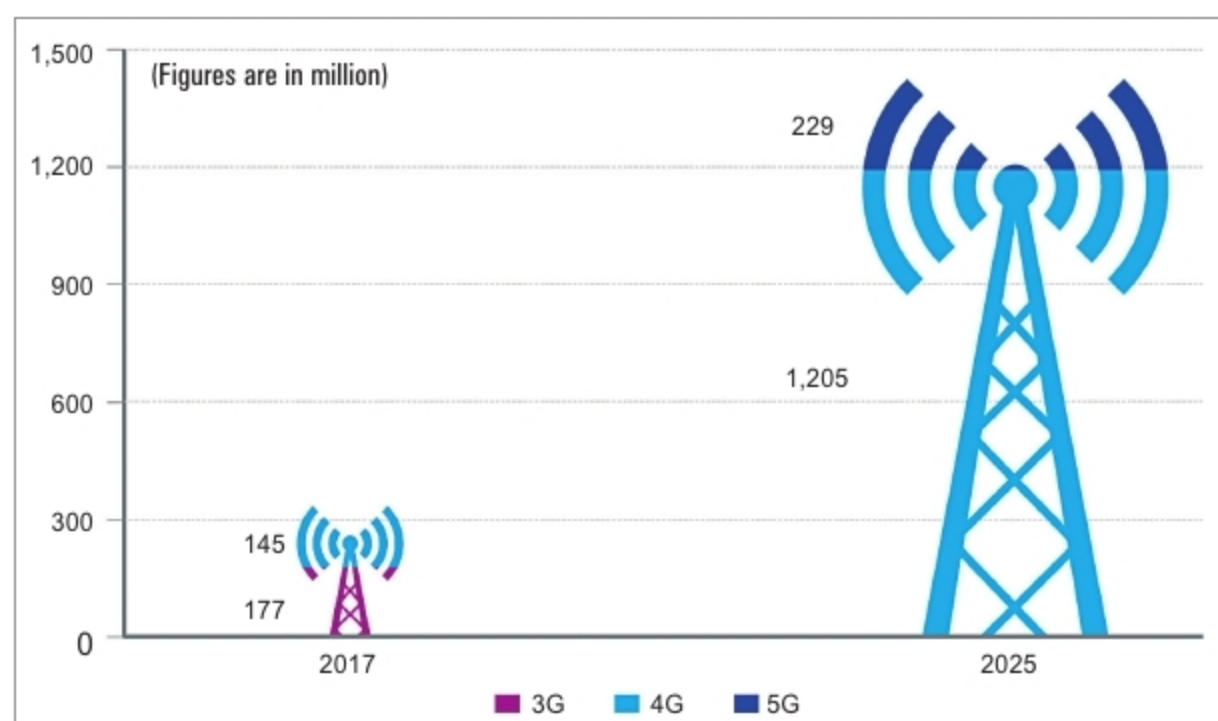


Figure 1: Projected growth in wireless broadband subscriber base, post 5G rollout (Image source: Edelweiss research)

avenues of employment and digitise the economy. 5G will also open opportunities for India to connect with global markets and take advantage of economies of scale. The 5G high-level forum is drawing up an action plan in this direction. 5G once implemented, will open up new avenues of revenue for service providers.

To justify the statement, Yash adds, “Covid-19 has definitely impacted the progress. There are now signs of coming out of it amidst fresh PE investments/FDI in the sector and regulatory support, but CAPEX spend as per IDC is expected to remain muted for Indian operators over 2019 to 2024. This is largely due to the fact that any increase in material change will

not immediately result in the CAPEX increase. Plus, Reliance Jio has already shown its interest in vendor-agnostic procurement, so the dynamics of the industry could shift from CAPEX-led investments to OpEx-led ones with the move towards open source and cloud-based workloads. So, once the auction takes place next year in an optimistic scenario, 5G test licenses will be followed by technology trials, industry formulation, global and regional cooperation with network sharing. So, the real commercialisation is only expected by 2022 or 2023 realistically.”

According to a report of COAI, the telecom industry, along with DoT, is leaving no stone unturned in making India 5G ready, and by next year, India

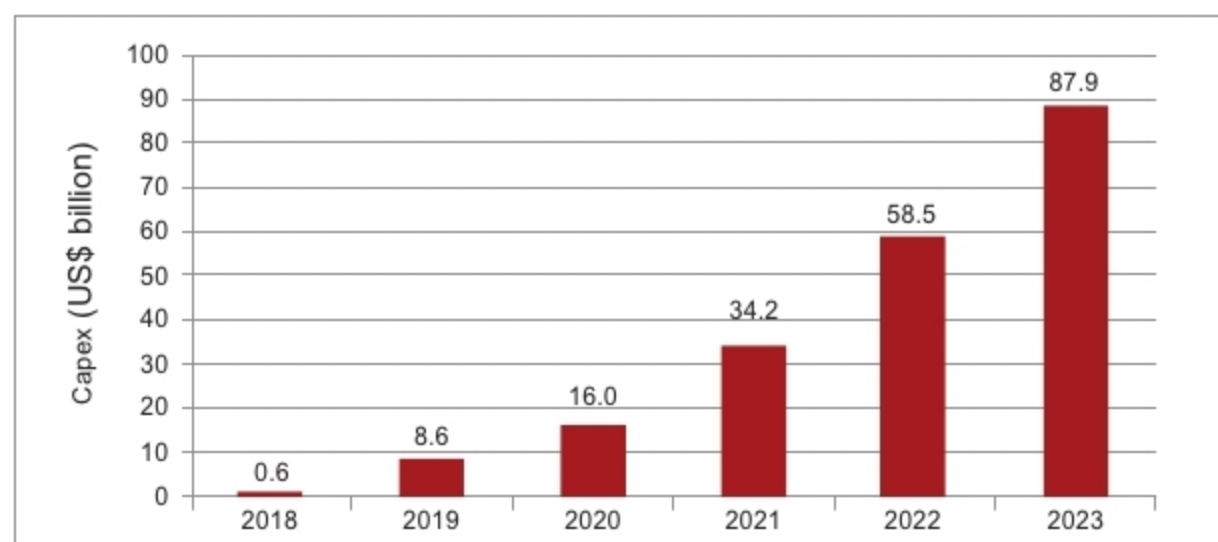


Figure 2: Forecast of annual global 5G capex by mobile operators (Image source: Heavy Reading)